PRAMAAN
प्रमाण · proof

GOOGLE CLOUD AI BUILDER CUP 2026 · SUBMISSION

CERTIFICATE OF EMBEDDED EMISSIONS · EU CBAM

THEME

Manufacturing:
Intelligent Operations & Industrial Efficiency

SUBMITTED BY

Team Innovative Warriors

BUILT FOR

Small steel-goods exporters · India & JAPAC

SOURCE CODE

github.com/JeyaanthAnandan/pramaan

CBAM proof from a phone photo.

An agentic Gemini copilot that turns photos of bills, invoices and handwritten logs into **verifier-ready EU CBAM data**. Every number traces back to the exact spot on the page it was read from, and shows what it is worth in € and ₹.

RESULTS · SAMPLE FACTORY, JAN–SEP 2026 — LUDHIANA

CHECK	REQUIRED	RESULT
Values read right, live Gemini	≥ 95 %	222/222
EC worked examples reproduced	all	8/8 PASS
Saved for the EU buyer, a year	–	€409,545



REPORT SIGNING KEY · CLOUD KMS · EC P-256

9451 0f5b bdf0 d336 3ef1 eef8 02c9 ff32

LIVE ON CLOUD RUN · SINGAPORE

pramaan-646454690406.asia-southeast1.run.app

BUILT ON

Gemini 3.8 · ADK 2.11 · KMS

EU buyers now pay for the carbon in every bolt. Small exporters can't prove theirs, so they are priced on penalty defaults.

DEFINITIVE PHASE

1 Jan 2026

EU importers now pay for the emissions embedded in steel imports[3]

INDIAN MSMES AT RISK

25–30 k

Fasteners, wire, pipes and more, mostly family-run[1]

EU SHARE OF EXPORTS

32–45 %

of India's steel exports go to the EU[1]

PRICE CUT AT STAKE

15–22 %

that exporters may need to absorb to keep EU orders[1]

COMPLIANCE COST / MSME

₹15–20 lakh

India is working on a scheme to reimburse 90 %[2]

MARK-UP ON EU DEFAULTS

+10→30 %

2026 → 2028+, on top of already conservative default values[3]

TODAY · A BOLT MAKER IN LUDHIANA — NON-CONFORMANCE

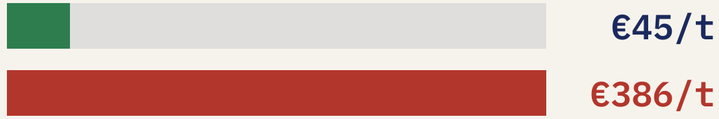
1. The German buyer writes: **send verified CBAM data per CN code**, or we use EU default values and you take the price cut.
2. The data exists, on paper: gas and power bills, rod invoices, a **Hindi production register**, a **handwritten Punjabi scrap log**.
3. A consultant quotes **₹15–20 lakh** and months of spreadsheets. Verifiers are booked out[4].
4. Nobody asks the rod supplier, the source of **90 % of the emissions**. Exporters in Vietnam, China, Korea and Indonesia face the same defaults.



[1] Business Standard via Vision IAS, 29 Jan 2026 · [2] The Indian Express via Business Standard, 30 Jun 2026 · [3] European Commission, CBAM Guidance No. 3, p. 115 · [4] EFDA via Eurometal

Defaults are a penalty that grows every year, and most quick estimates get the maths wrong.

BUYER'S CBAM COST, 2026 IMPORTS · CARBON-STEEL BOLTS, CN 7318 15 82



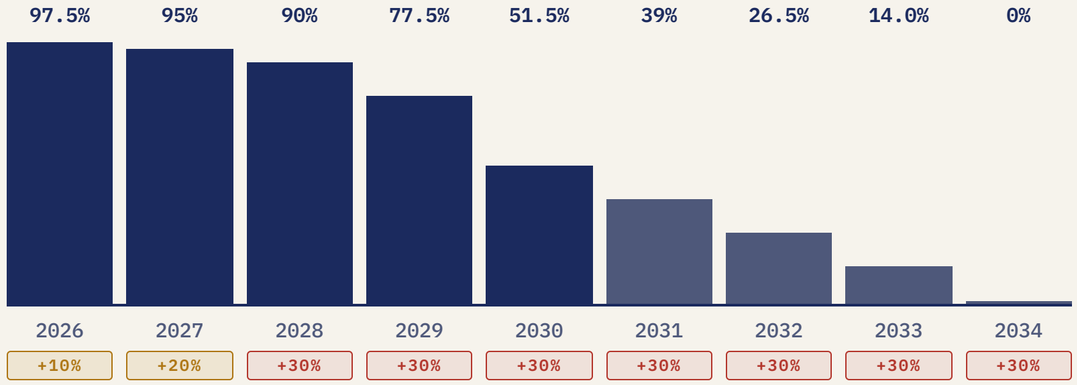
€341 /t on the table, **₹36,994 per tonne**, which is **≈ €409,545 a year** at the sample factory's volumes

2026 certificate price: Q1 €75.36 · Q2 €75.28 · Q3 €82.32, weighted by export quarter = **€77.85**.

THE TRAP · STAINLESS BOLTS, ROD ON INDIA'S DEFAULT

On current data the buyer would pay **€576/t**, more than the EU default (**€402/t**). Pramaan shows the cheaper path, and the step that changes it: verified rod data → **€97/t**.

FREE ALLOCATION STILL DEDUCTED (CBAM FACTOR) · DEFAULT MARK-UP



Every year less is deducted, so every tonne above the benchmark costs more, while the default mark-up triples.

MYTH-BUSTER

"Buyers pay **2.5 %** in 2026" is wrong. The Commission's formula subtracts a benchmark-based free allocation, so everything above the benchmark is payable: EC Example 1 has **3,690 certificates on 15,670 t** embedded, **23.5 %**.

CERTIFICATES ON SALE
1 Feb 2027

FIRST DECLARATION DUE
30 Sep 2027
for goods imported in 2026

Schedule: EC Guidance No. 4, Table 2-1, p. 13 · mark-ups: Guidance No. 3, p. 115 · prices: EC certificate-price page · example: Guidance No. 5d, p. 44 · dates: Reg. (EU) 2023/956 as amended by 2025/2083 · costs: Pramaan sample factory (fictional)

Photos in, verifier-ready CBAM data out, with every number traced to its pixel.

IN · 24 PHONE PHOTOS

- Gas & power bills
- Rod invoices
- Supplier statements
- Hindi production register
- Handwritten Punjabi scrap log
- Export invoices

→

STEP 1 · CAPTURE AGENTS LIVE

"Add your documents."

Phone camera or upload: JPG, PNG, HEIC, PDF. The agent timeline shows each Gemini step as it runs. Laptop? A QR code continues on the phone.

→

STEP 2 · CLOSE GAPS YOUR TAP

"What's missing?"

Green: two readers agree. Amber: needs your tap. Red: missing or on defaults. Answer by voice in Hindi, Punjabi or Tamil.

→

STEP 3 · REPORT 99 % READY

"Here's your proof."

The verdict first, then **What to do next** with € attached. SEE per CN code. The Evidence Lens on every number.

→

OUT · TO THREE PARTIES

EU buyer link
summary only, no plant data

Verifier pack
signed, with a spot-check sample

Supplier letter EN / हिंदी
with a no-account upload link

THE THEME, LITERALLY · "TRANSFORM OPERATIONAL DATA INTO MEANINGFUL INSIGHTS, RECOMMENDATIONS, OR ACTIONS"

OPERATIONAL DATA
Bills, registers, scrap logs
Photographed on the shop floor, in four languages and two scripts, some of it handwritten.

INSIGHTS
SEE per product, and where it comes from
The rod is 90 % of a carbon bolt's emissions. June's furnace ran 18.5 % hot.

RECOMMENDATIONS
"What to do next", priced
Ask the stainless rod supplier for verified data: –€306/t for the buyer.

ACTIONS
Letter, link, levers
Supplier letter on WhatsApp, buyer link, signed verifier pack, operations levers.

OWNER / ACCOUNTANT · LUDHIANA
Phone-first, Hindi or Punjabi. Needs: what to upload, what's missing, what it costs.

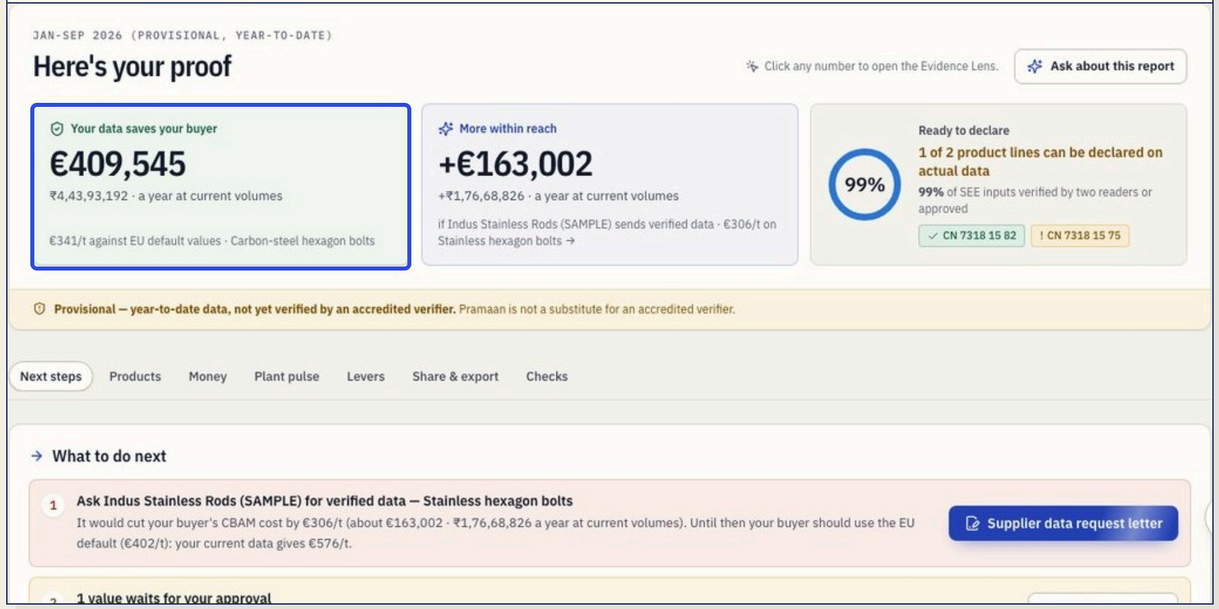
EU IMPORTER · THE DECLARANT
Needs SEE per CN code and the data status, without the plant's secrets.

ACCREDITED VERIFIER
Needs source records and controls: a signed pack and a sample to check.

ROD SUPPLIER
One clear request, one upload on a phone, no account, sees only its own reading.

The report answers first: what the data is worth to the buyer, and what to do next.

EXHIBIT A · REPORT · LIVE GEMINI RUN, 13 OCT 2026 — PRAMAAN…RUN.APP



99 % of SEE inputs are verified by two readers or approved, yet only **1 of 2 lines** can be declared on actual data: the stainless rod is still on its default. Pramaan says so instead of rounding up.

1 OF 2 LINES READY
STAINLESS ROD ON DEFAULT

PRODUCT LINE	CARBON BOLTS	STAINLESS BOLTS
CN code	7318 15 82	7318 15 75
SEE, t CO_2 per t	2.040	8.889
Rod's share of SEE	90 %	98 %
Buyer pays, 2026	€45/t	€402/t
Basis	ACTUAL · VERIFIED	EU DEFAULT

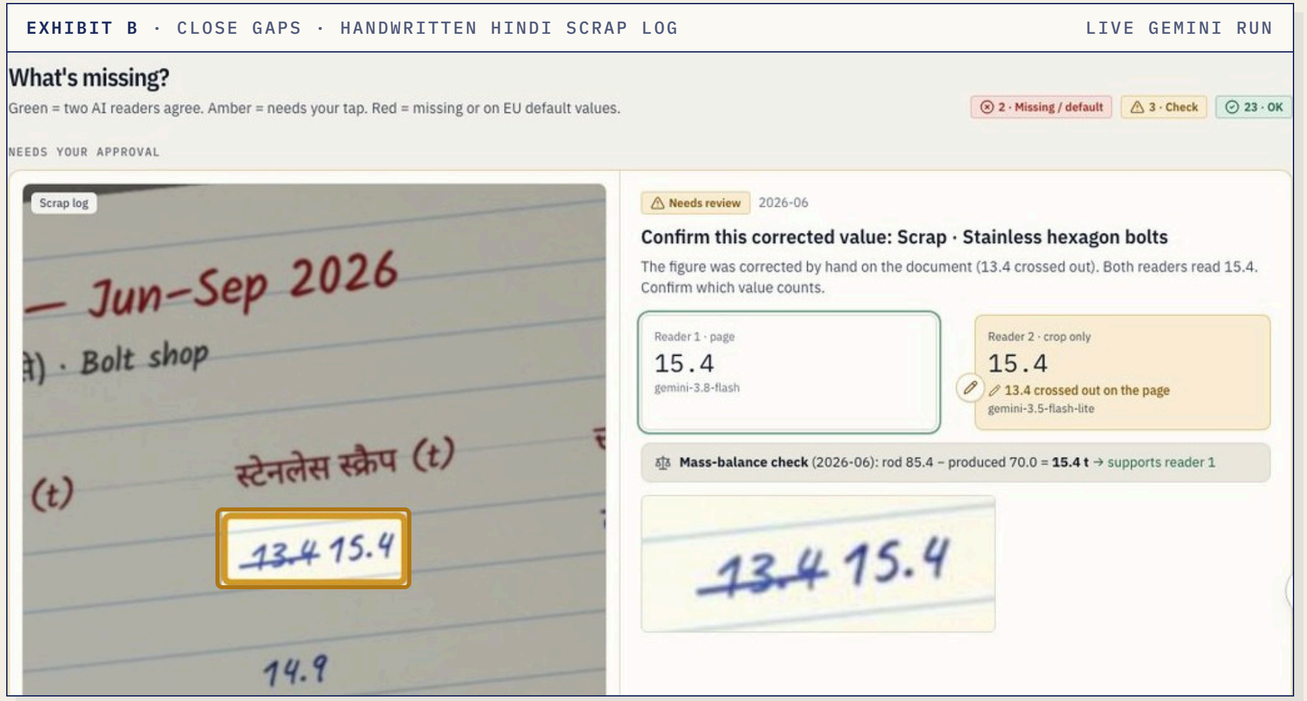
WHAT TO DO NEXT · TOP OF THE REPORT

1. **Ask Indus Stainless Rods for verified data.** −€306/t for the buyer, ≈ €163,002 a year.
2. **Approve one value:** a scrap figure corrected by hand on the page.
3. **Send the carbon-steel data to the buyer:** €341/t below the EU default.

SEE 2.040 equals EC Guidance 5d Example 3 at full precision (2.0399). Stainless: rod on India's default value 6.480 + 10 %, so the cheaper path for the buyer is the EU default (€402/t, vs €576/t on current data).

Two different Gemini models read every value. The second one sees only the crop.

EXHIBIT B · CLOSE GAPS · HANDWRITTEN HINDI SCRAP LOG — LIVE GEMINI RUN



VALUES READ	READERS AGREE	SENT TO A PERSON	24 DOCS → REPORT
222	99.5 %	1	96.8 s

1. **Gemini 3.8 Flash** classifies each page and reads every required value with its box_2d: JSON schema, high media resolution, Hindi, Punjabi and English on one page.
2. Each box is cropped with 15 % padding. **Gemini 3.5 Flash-Lite** re-reads **only the crop** and never sees the first answer.
3. Agree within 0.5 % → `VERIFIED · 2 READERS`. Disagree, or corrected by hand on the page → `NEEDS YOUR TAP`. A tap → `APPROVED`, with who and when.
4. On screen: **13.4 struck out, 15.4 written.** A deterministic mass balance, rod 85.4 – produced 70.0 = **15.4 t**, shows which reading closes. The owner decides.

WHAT THE MODEL DOES THAT OCR AND RULES CAN'T

Reads crossed-out handwriting in Devanagari and Gurmukhi, tells the GCV from the meter reading, files each value to its month and product line, and returns a box we can crop and prove.

Numbers from backend/eval/report.md, live run 13 Oct 2026 (gemini-3.8-flash + gemini-3.5-flash-lite) on the 24-document photographed sample set with 222 ground-truth fields.

Click any number: its formula, every value behind it, and the exact pixel it came from.

REPORT	**2.040** t CO_2/t · SEE, carbon-steel bolts
↓	
FORMULA	$AttrEm/AL + \Sigma\ m_i \times SEE_i$ = **0.196** own process + **1.844** wire rod
↓	
INPUTS	**38 values from 12 documents**, 1 official value, 3 stated assumptions
↓	
SOURCE	March gas bill: **803.543 MMBtu**, boxed on the photo 2 READERS

THE RULE

The LLM never computes a compliance number. A Decimal engine does: 380 nodes, each tied to a photo region, an official EC file (with its SHA-256) or a stated assumption. No regulatory value is typed by hand.

Verifier spot-check: Pramaan draws the sample a verifier would (every value a person touched, the supplier's data, then the rest by weight). Each one is shown as its crop, so a year of paper becomes a dozen crops checked in minutes.

EXHIBIT C · THE LENS

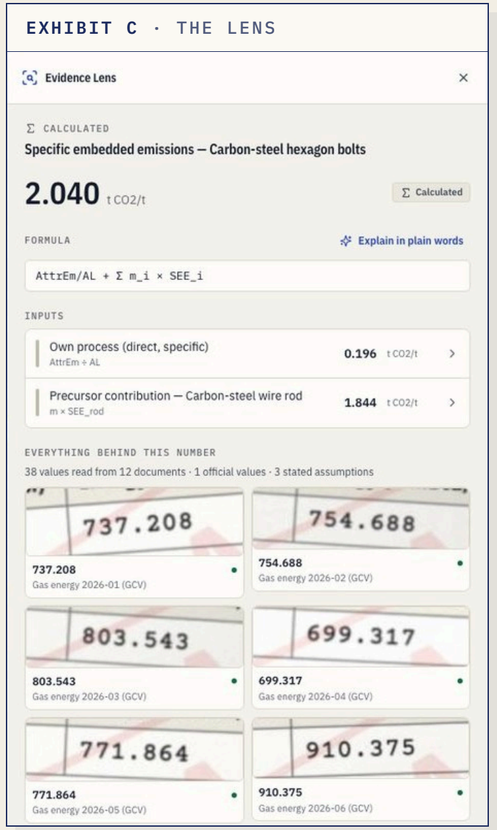
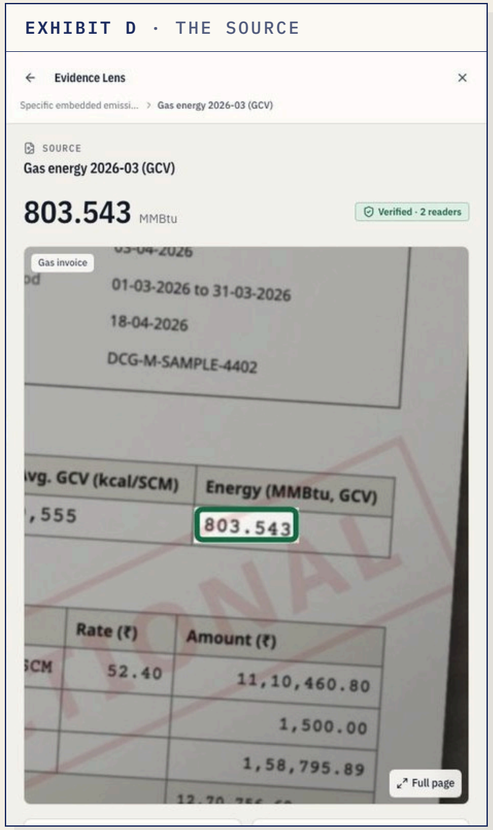


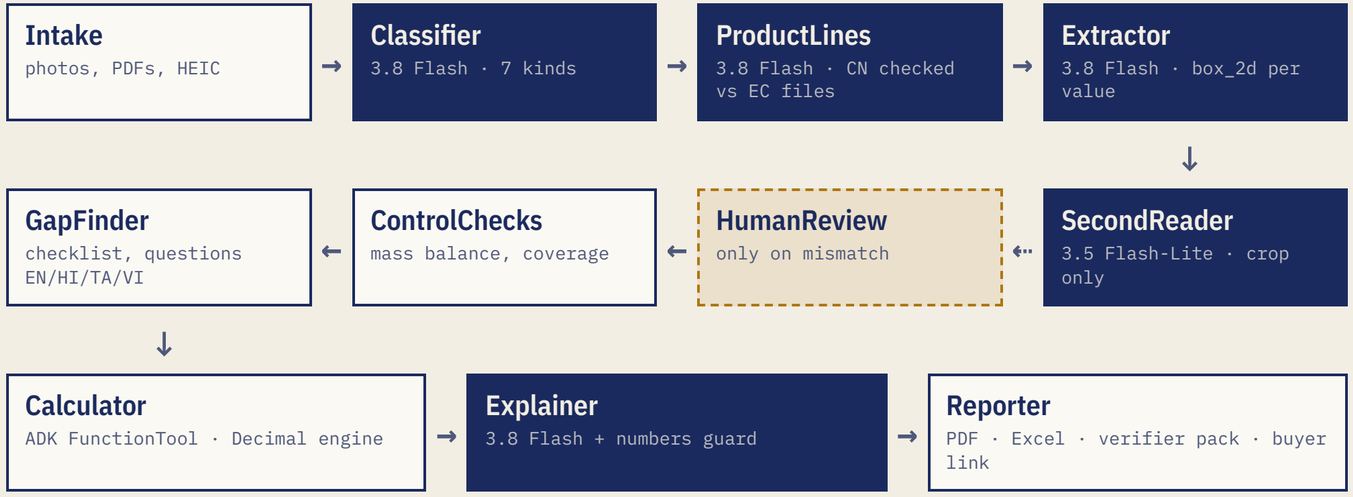
EXHIBIT D · THE SOURCE



An ADK agent graph does the work. Gemini reads and explains; code computes and checks.

ADK 2.11 WORKFLOW GRAPH · ROUTE-BASED BRANCHING

GEMINI · CODE · PERSON



google-genai **Interactions API**: JSON-schema output, a thinking level per agent, high media resolution for documents. Only Gemini 3.x is allowed, and a test rejects anything else. Every node streams to the live timeline with agent, model, time and tokens.

AGENT SCENARIOS IN CI	…AND AS AN ADK EVAL SET	HARD LIMITS
skewed handwritten log → HumanReview; Hindi-headed register; a bill with two periods	missing supplier statement → default + lever; mismatched unit → review	every Gemini call has a time limit and retry; a backup request after 30 s, first answer wins

NUMBERS GUARD · EVERY NUMBER TIED TO ITS OWN SOURCE

WHERE	THE CHECK	IF IT FAILS
Ask about this report	each number cites the fact `[F:id]` that shows it	one repair, then withheld
Rules copilot	printed on the EC page cited in that sentence	the pages are shown instead
Narrative	only its template's numbers; keeps the SEE	template text
Call Pramaan	spoken numbers are report facts or tool answers	flagged; it corrects itself aloud

PASSED 1ST TIME	REPAIRED	WITHHELD
11 /12	1	0

Live questions in English, Hindi, Tamil and Vietnamese. Text typed by owners and suppliers, and text written on a photo, goes into prompts as data, never as instructions.

Talk to your report in five languages. Every number it says is checked.

EXHIBIT E · A CALL IN HINDI



ENGLISH | हिंदी | ਪੰਜਾਬੀ | தமிழ் | TIẾNG VIỆT

1. **Gemini 3.8 Live** on Agent Platform answers out loud in **1.1–2.3 s**, and the caller can interrupt.
2. **Four tools run in our code:** `explain_number` · `open_evidence` · `search_guidance` · `open_items`. It cannot compute, approve or send.
3. The server transcribes the agent's own voice and matches every number to the report. A number that isn't in it is flagged, and **the agent corrects itself aloud**.
4. Credentials stay on Cloud Run: 16 kHz audio over a WebSocket, a 5-minute cap, no recording kept.

FROM A LIVE CALL · HINDI — MATCHED TO THE REPORT

Caller: मेरे खरीदार को हर साल कितनी बचत होगी? (How much will my buyer save a year?)

Pramaan answers aloud in Hindi; the card `409,545 EUR/year · 4,43,93,192 INR/year` appears and opens its proof.

EXHIBIT F · /TRY · ONE PHOTO, TWO MODELS — BUNDLED REPLAY



Try one photo streams the three reading steps on a single page in seconds: boxes appear, then turn green or amber. A judge's own bill is **read in memory and never stored**.

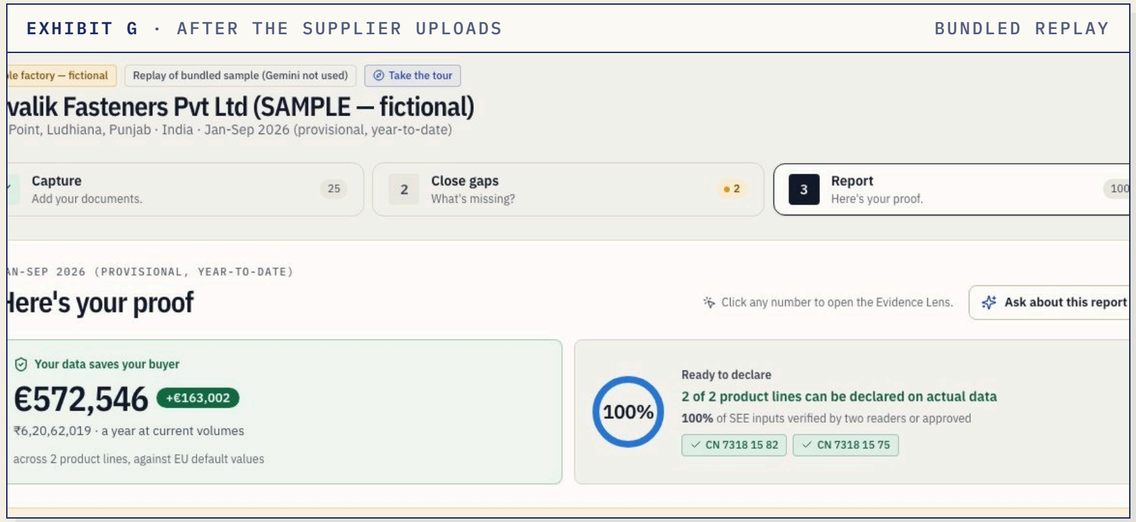
READER 1	READER 2	CAPPED
3.8 Flash classifies, reads and boxes	3.5 Flash-Lite, each crop, blind	20 live tries per address an hour

Tested live 13 Oct 2026: spoken questions in English, Hindi, Tamil and Vietnamese, typed in Punjabi; answers in the caller's language with the right figures and no false warnings. gemini-3.8-live is served from us-central1.

Recommendations with the money attached, and actions that move it.

ACTION · CLOSE THE LOOP WITH THE ROD SUPPLIER

EXHIBIT G · AFTER THE SUPPLIER UPLOADS — BUNDLED REPLAY



STAINLESS, BUYER €/T	SAVED A YEAR	LINES READY
402 → 97	€572,546	2 of 2

The letter (EN / हिंदी) carries an upload link, sent on WhatsApp. The supplier uploads on a phone with no account; the same agents read it; the exporter's open report moves by itself. **The supplier never sees plant data; the buyer never sees the supplier's.**

INSIGHT · PLANT PULSE: THE SAME GAS BILLS SHOW HOW THE FURNACE RAN

EXHIBIT H · GAS PER TONNE, MONTH BY MONTH



JUNE RAN HOT	GAS ABOVE MEDIAN	WASTED
+18.5 %	₹2,00,905	7.6 t CO_2

Operations levers on the same engine: EC Example 2 rod data plus electric heat treatment take the stainless bolts from SEE 8.889 to **2.174** and the buyer's cost from €402 to **€81/t**. Cutting scrap from 15 % to 10 % saves **111 t of rod a year**.

Sample factory (fictional), Jan–Sep 2026. Exhibit G replays the bundled supplier statement ("Open as the supplier"); live, the same flow runs Gemini on the supplier's upload.

Reports nobody can quietly change, and numbers the AI can't invent.

CHANGE LOG · EACH ENTRY CARRIES THE SHA-256 OF THE ONE BEFORE

Value read
by a run

Approved
or changed by hand

Spot-check
verifier's mark

Report issued
PDF · pack · link

RECEIPT ON EVERY REPORT · SIGNED IN CLOUD KMS — ECDSA P-256

Headline figures + how each value was confirmed (two readers / approved / **changed by hand** / waiting) + SHA-256 of the calculation tree, the values, **every source photo**, the refdata, the log head and each file issued. **The private key never leaves KMS.**

```
$ openssl dgst -sha256 -verify receipt/public_key.pem \
    -signature receipt/statement.sig receipt/statement.json
Verified OK
```

`/r/{id}` checks the signature in the reader's own browser (WebCrypto) and compares a dropped-in PDF byte for byte. Key fingerprint **`9451 0f5b bdf0 d336 3ef1 eef8 02c9 ff32`**, published in the README.



RISK REGISTER

RISK	GATE
Hallucinated numbers	LLM never computes; numbers guard on text and speech
Misread documents	blind crop re-read, mass-balance checks, a person for hand corrections
Wrong regulation values	parsed from EC files with SHA-256 manifest; CI fails on drift; 8 golden tests
Prompt injection	document and typed text enter prompts as data, never instructions
Plant data leaking	passcode (hash only), HttpOnly cookie; buyer sees a summary, supplier sees its own reading; delete for good
Abuse and cost	per-address caps; uploads validated; 5-min calls; nothing stored on /try
Over-claiming	"provisional until verified"; not a substitute for an accredited verifier

ANSWERING THE VERIFICATION BOTTLENECK

The verifier's marks go on the same hash chain, print as a record and ship in the pack as `spot_check.csv`. Site visits stay the verifier's job; the desk review gets fast.

One Cloud Run service, Gemini on Agent Platform, and no API keys in production.

WHO USES IT

Exporter
phone PWA · EN/HI/TA/VI

Rod supplier
upload link, no account

EU buyer
summary link

Verifier
signed pack, spot-check

⇄

CLOUD RUN · PRAMAAN · ASIA-SOUTHEAST1

1 CONTAINER

FastAPI + SSE + React PWA
one service, one cold start · min-instances 1 for judging

ADK 2.11 Workflow
Classifier · ProductLines · Extractor · SecondReader · Explainer

Decimal engine
refdata + SHA-256 manifest

Ask + Rules copilot
facts · EC pages

Try one photo
SSE, nothing stored

Call Pramaan
WebSocket audio

⇄

GOOGLE CLOUD

Agent Platform · global
gemini-3.8-flash · gemini-3.5-flash-lite

Agent Platform · us-central1
gemini-3.8-live (voice)

Firestore · Cloud Storage
native mode · private bucket

Cloud KMS
EC P-256 report signing

Logging · Trace · IAM
least-privilege service account

EXHIBIT I · RUN TRACE



Every run leaves a receipt: agent, model, time, tokens.

BUILD & DEPLOY

Cloud Build + Artifact Registry; `deploy.sh`: golden tests → `gcloud run deploy` → smoke test (sample, buyer link, receipts trusted)

CI ON EVERY PUSH

ruff · mypy · pytest (245) · golden tests · refdata drift · gitleaks · logged-out UI smoke on desktop and a 360 px phone

SCALES BY DESIGN

Stateless container, runs followed across instances via Firestore; 15 visitors at once, 0 errors in the load test

Project copilot-manufacturing · scripts/gcp_setup.sh creates the APIs, Firestore (asia-southeast1), a private bucket, the service account and the KMS key.

Checked against the Commission's own worked examples, and measured on live Gemini.

GOLDEN TESTS · EC GUIDANCE 5D (14 AUG 2026) · ALSO LIVE ON THE EC PROOF PAGE **8 / 8 PASS**

ID	EC EXAMPLE	EXPECTED	PRAMAAN
G1	Ex. 3 · carbon screws & nuts · p. 53	SEE 2.039	2.0399 ✓
G2	Ex. 3 · high-alloy screws & nuts · p. 53	SEE 2.371	2.3707 ✓
G3	Ex. 1 · SEFA rails · pp. 43–44	1.261 → 1.198	1.2616 → 1.1985 ✓
G4	Ex. 1 · certificates, rails · p. 44	3,690	3,690.0 ✓
G5	Ex. 2 · certificates, pipes · p. 51	95.9	95.93 ✓
G6	Ex. 1 · SEFA sheets & bars · p. 43	1.248 · 1.242	1.2486 · 1.2426 ✓
G7	Ex. 3 · SEFA screws · p. 54	1.499 · 1.159	1.4992 · 1.1587 ✓
G8	Benchmarks & gas EF, re-read from official files	0.057 … 56.1	exact ✓

We also found **two inconsistencies in the EC's own examples** (FeCr SEE 2.280 vs an implied 2.820; pipes 1.783 vs 1.803), logged and kept out of the tests.

EXTRACTION EVAL · LIVE RUN, 13 OCT · 24 PHOTOS, 222 FIELDS

METRIC	RESULT	TARGET
Field accuracy	100 %	≥ 95 % ✓
Region hit, IoU ≥ 0.5	98.6 %	–
Two-reader agreement	99.5 %	–
24 documents → report	96.8 s	< 120 s ✓

A FACTORY THE APP HAD NEVER SEEN

158 of 162 values right, **0 wrong**; results match an independent calculator to **6 decimals**.

HONEST LEARNING CURVE · FIELD ACCURACY, LIVE RUNS



AUTOMATED TESTS	AXE · WCAG A/AA	LIGHTHOUSE · MOBILE	FIRST LOAD · GZIP	STATIC CHECKS
245	0 violations	91·100·100·100	151 KB JS	ruff·mypy·gitleaks

backend/tests/test_golden_ec.py · backend/eval/report.md · docs/TEST_REPORT.md (independent new-factory test, 14 Oct) · tolerance ±0.002 t CO_2/t (EC tables round intermediates).

Built for every small exporter between a CBAM bill and a phone camera.

IMPACT MODEL

PER FACTORY (SAMPLE)

€409,545 /yr

kept off the buyer's CBAM bill today, +€163,002 within reach: margin the exporter need not give away

REACH

25–30k MSMEs

in India alone; the same defaults hit Vietnam, China, Korea, Indonesia

SCALE PATH

Proposed downstream extension (~180 product categories from 2028), UK CBAM from 2027, then aluminium, cement and fertilisers on the same engine



DIFFERENTIATION

	CONSULTANT + SPREADSHEET	GENERIC AI CHAT / OCR	PRAMAAN
Reads photos, handwriting	by hand	yes, unchecked	**two blind readers**
Number → source	binder of copies	no	**the pixel, signed**
Who does the maths	a spreadsheet	the LLM	**Decimal engine, 8/8 EC**
Buyer's € and next step	rarely	no	**on every line**
Languages	English	varies	**EN·HI·PA·TA·VI, voice**
Cost, time	₹15–20 lakh, months	low, no audit trail	**minutes, on existing bills**

REAL TODAY

Live Gemini on Agent Platform, Cloud Run, KMS-signed receipts, refdata parsed from the EC's files, own-factory projects

SAMPLE

The factory and its 24 documents are fictional and watermarked; "Load sample factory" serves a captured live run, and says so

PROPOSED 90-DAY PILOT

30 — **10 fastener exporters** in Ludhiana with their EU buyers, on their own bills, with consent.

60 — **One accredited verifier** runs the spot-check on real verifier packs.

90 — Measure **hours to verifier-ready**, € per tonne moved, desk-review time.

Already built for it: private own-factory projects, product lines proposed from invoices, supplier upload links, the verifier spot-check, and delete-for-good.

THE ASK

Shortlist Pramaan: it turns the paper a small factory already keeps into proof Europe accepts, and puts the money on every line.